

Operating Systems

Menuet OS (Memory Management)

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Memory Management:

MenuetOS uses a simple but effective memory management system. It employs a flat memory model where all memory is accessible through a single flat address space. The operating system manages memory through a system of memory pools that are allocated at boot time.

When the operating system boots up, it creates a number of memory pools of fixed sizes that are used to allocate memory for various system components and applications. These memory pools are managed by a central memory manager that keeps track of which memory blocks are in use and which are free.

A memory pool is a contiguous block of memory that is reserved by the operating system for a specific purpose

The memory manager provides a simple interface for allocating and freeing memory blocks. When an application requests memory, the memory manager searches for a free block of the appropriate size from the available memory pools. If a suitable block is found, it is marked as in use and returned to the application. When the application is finished with the memory, it returns it to the memory manager, which marks the block as free and adds it back to the pool of available memory.

MenuetOS also supports virtual memory, which allows applications to access more memory than is physically available. However, the implementation of virtual memory in MenuetOS is limited and does not support features such as demand paging or memory protection.

Ex:

An application requests 100 bytes of memory from the operating system. The memory manager will search for a free memory block of the requested size in the available memory pools. If a suitable block is found, it is marked as "in use" and returned to the application. If no suitable block is found, the memory manager can allocate a new block of memory from the available memory pool.

When the application is finished with the allocated memory, it returns it to the memory manager, which marks the block as "free" and adds it back to the pool of available memory. This way, the memory manager can reuse freed memory blocks for future requests, minimizing memory fragmentation and maximizing the available memory for the operating system and applications.

References:

- The official MenuetOS website: <http://menuetos.net/>
- The MenuetOS Wiki: http://wiki.menuetos.net/Main_Page
- The MenuetOS Developer's Guide:
<http://menuetos.sourceforge.net/developersguide.html>